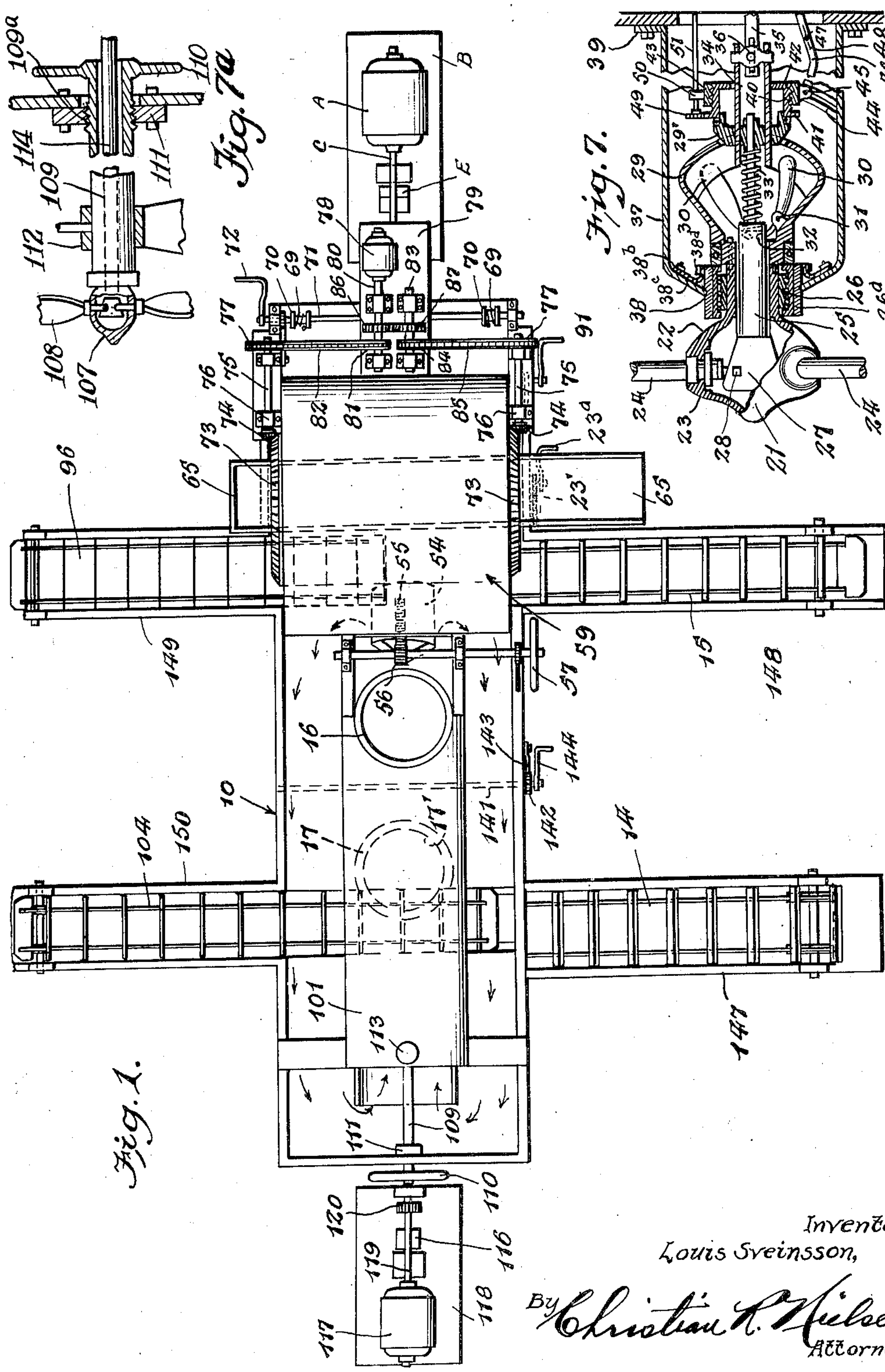
March 30, 1937. L. SVEINSSON 2,075,593

APPARATUS FOR CLEANING COAL OF FOREIGN MATTER

Filed April 16, 1934 4 Sheets-Sheet 1



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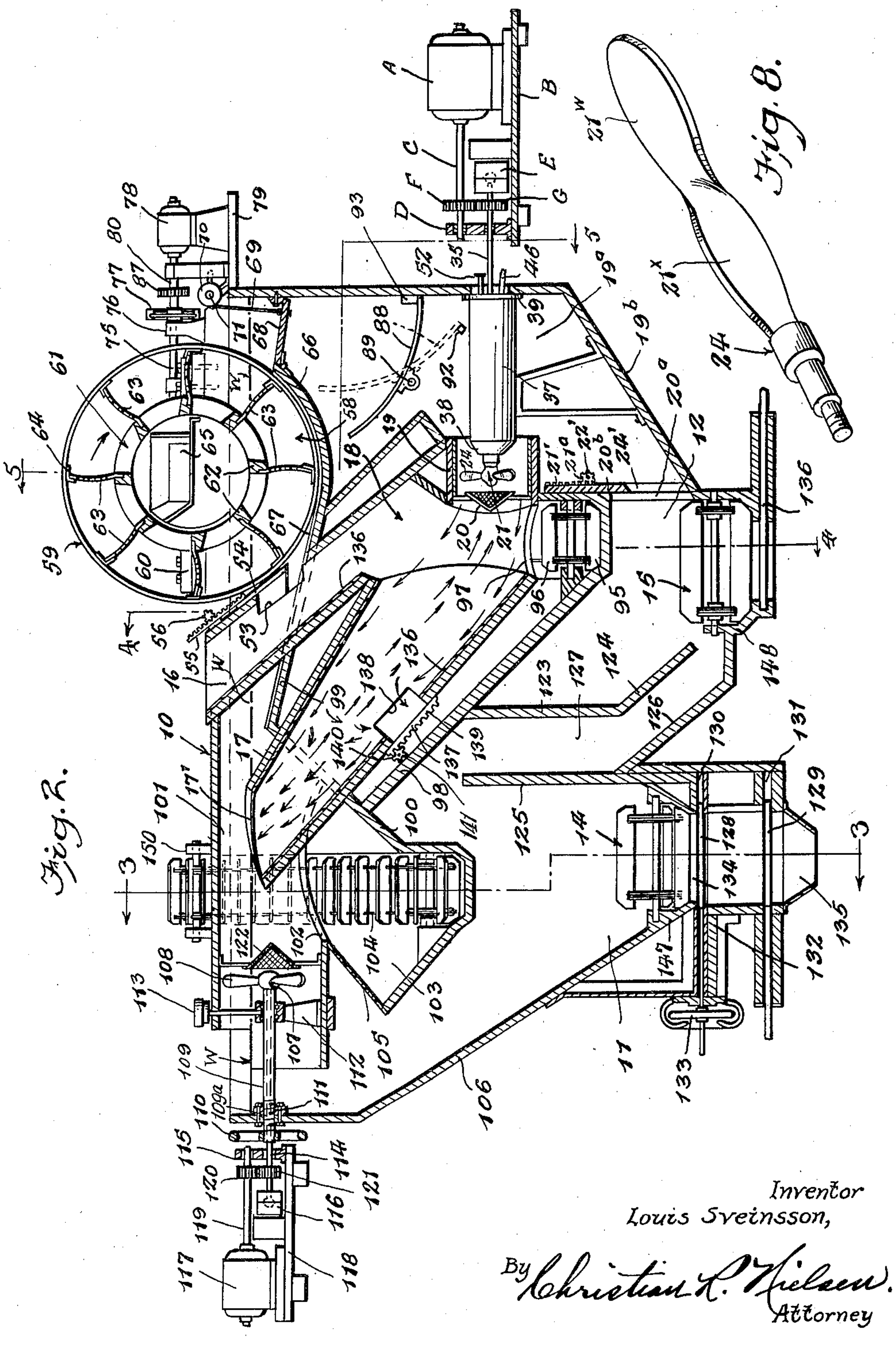
APPARATUS FOR CLEANING COAL OF FOREIGN MATTER

Filed April 16, 1934 4 Sheets-Sheet 2



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APPARATUS FOR CLEANING COAL OF FOREIGN MATTER

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Fig. 3.

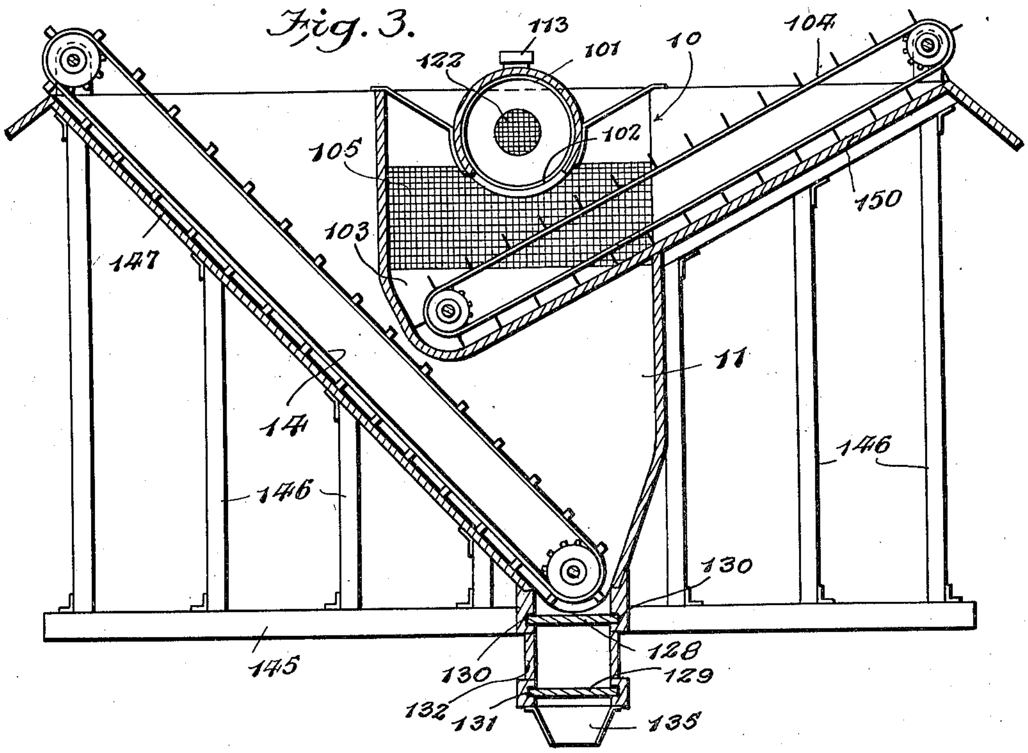
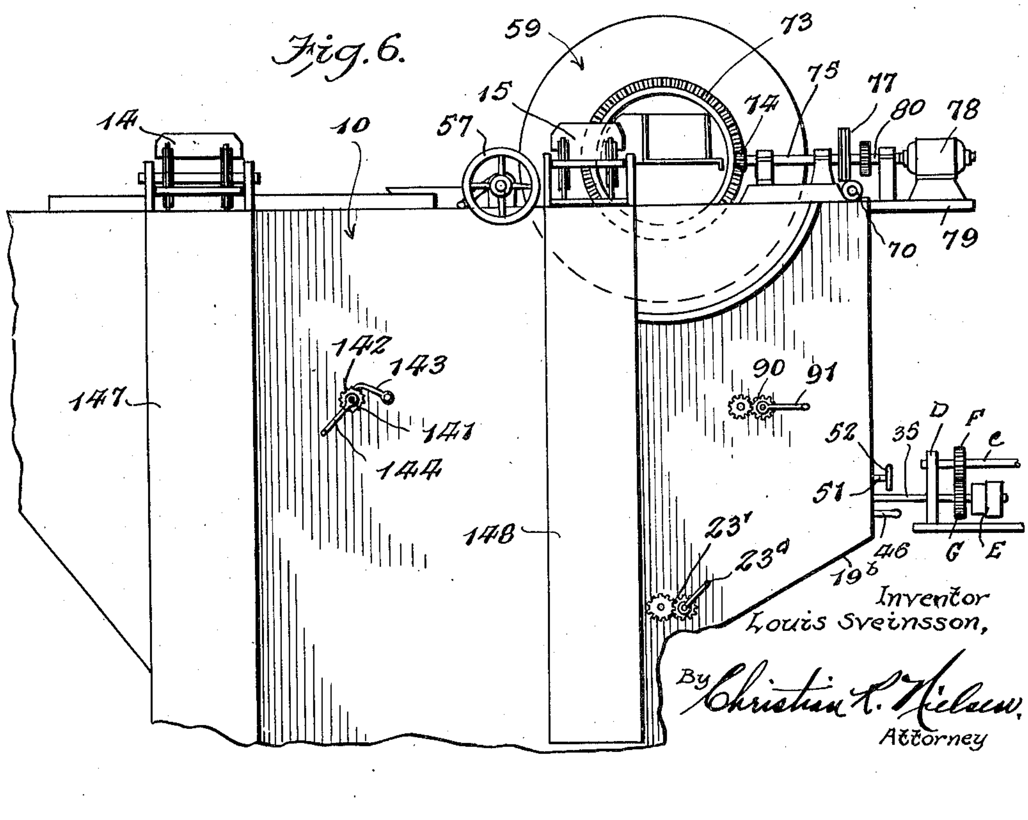


Fig. 6.



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APPARATUS FOR CLEANING COAL OF FOREIGN MATTER

Filed April 16, 1934 4 Sheets-Sheet 4

Fig. 4.

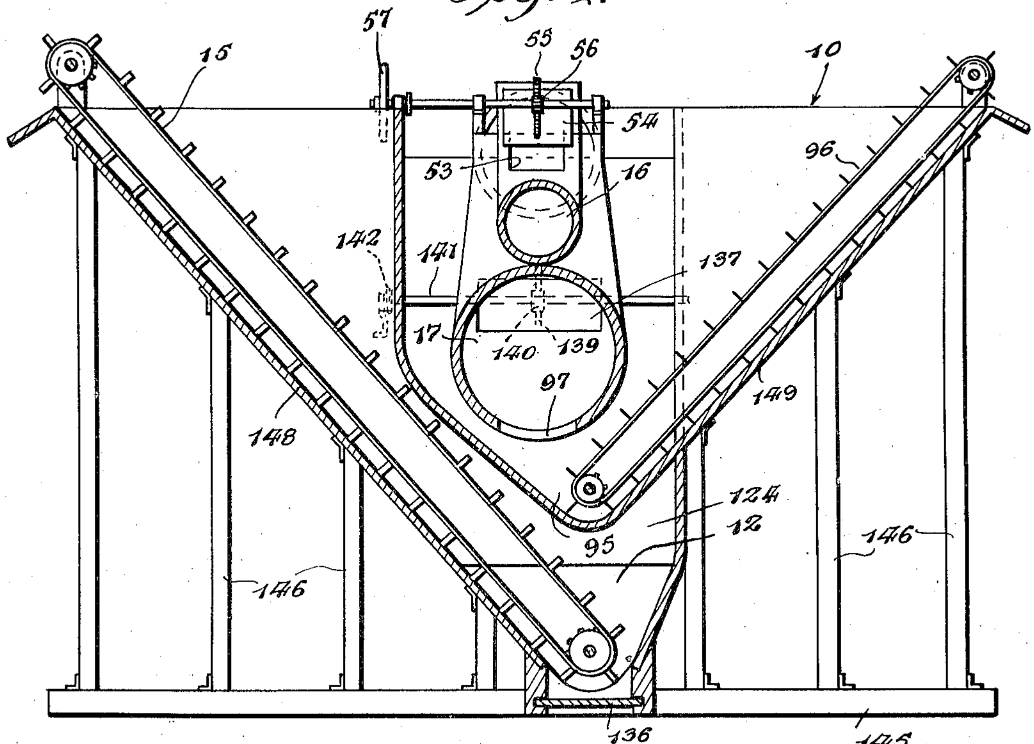
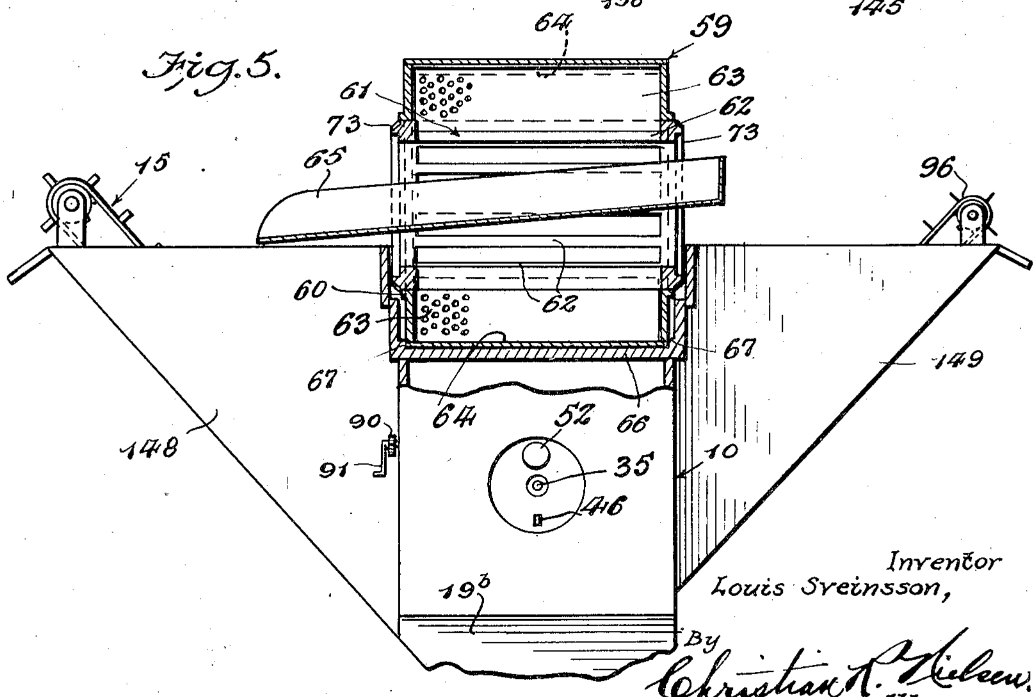


Fig. 5.



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Patented Mar. 30, 1937

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UNITED STATES PATENT OFFICE

2,075,593

APPARATUS FOR CLEANING COAL OF FOREIGN MATTER

Louis Sveinsson, Shamokin, Pa.

Application April 16, 1934, Serial No. 720,862

5 Claims. (Cl. 209—173)

This invention relates to an apparatus for the separation of foreign matter from coal such as clum, bank coal and river coal.

It is a special object of the invention to provide an apparatus of simple construction and in which water is caused to circulate in a spiral course through the aggregate being cleaned in such manner that a buoyant effect is produced, thereby facilitating the efficient separation of coal from undesired débris.

It is a particular object of the invention to provide a novel means for effecting simultaneously a reverse whirl flow of water through the coal being cleaned, thereby insuring thorough separation of the material under treatment.

It is also an object of the invention to provide a structure wherein a combined water and air current through the aggregate bulk under treatment may be carried out, whereby floating particles may be positively separated from the bulk.

It is also an object of the invention to provide a novel means of combining air and water wherein influential buoyancy is caused to float porous cinders so as to separate them from the coal aggregate.

Additional objects, advantages and features of invention will be apparent from the following description and accompanying drawings forming a part of this application, wherein

Figure 1 is a plan view of the apparatus embodying the invention.

Figure 2 is a longitudinal sectional view thereof.

Figure 3 is a cross section on the line **3—3** of Figure 2.

Figure 4 is a cross section on the line **4—4** of Figure 2.

Figure 5 is a cross section on the line **5—5** of Figure 2.

Figure 6 is a fragmentary side elevation of the apparatus illustrating, particularly, the drive for the pick-up drum.

Figure 7 is a detail sectional view of one of the propellers.

Figure *7a* is a detail sectional view of another propeller employed in the apparatus.

Figure 8 is a detail perspective view of one of the propeller blades.

In carrying out my invention I provide a main tank **10** of a water-tight nature, open at its top and in its base there are provided a pair of bins **11** and **12**, within each of which endless conveyors **14** and **15** respectively are mounted, as will be explained in greater detail hereinafter.

Opening upon an upper portion of the tank **10** a filling chute **16** is arranged, as clearly shown in Figures 1 and 2. This chute **16** extends downwardly into the tank for approximately one-half of its depth and is joined adjacent the base thereof by a diagonally disposed chute **17**, the open upper end **17'** being positioned a short distance below the top of the tank. The juncture of the chute **16** and **17** defines a well **18**.

From the description thus far set forth, it will be apparent that the coal to be cleaned is led into the apparatus through the chute **16**, settling in the well **18** by action of gravity, and it is at this point that the most effective cleaning operation occurs, as will now be explained.

Communicating with the well **18** there is a tunnel **19**, disposed at right angles thereto, the port forming communication with the well being suitably screened at at **20**. The tunnel **19** has revolubly housed therein a propeller **21**, the blades of which are adjustable to vary the "push" or "pull" of water through the coal being cleaned and will be dealt with in detail in the description of the operation of the apparatus. The propeller **21** may include any desired number of blades, although in the present instance, three have been illustrated and while the details of the propeller structure forms the basis of a copending application filed May 29, 1934, Serial Number 728,155, a general description of the construction will be given. Briefly, the blades adjacent the hub of the propeller are given an angular set as at **21***x* opposite that of the extremities of the blades as at **21***w*, as clearly shown in Figure 8, the purpose of which will be apparent as the description proceeds.

It should be noted that the tunnel **19** opens into a compartment **19***a* defined by one end and inclined bottom wall **19***b* of the tank and a vertically disposed apertured partition wall **20***a*. The bottom wall **19***b* is inclined so as to more readily feed debris to the conveyor **15**, as well as to permit free circulation of water through the tank **10**. However, means are provided for controlling the passage of water and débris from the compartment by the provision of a sliding valve **20***b*. The valve **20***b* includes a rack bar **21'** with which there is meshed a gear **21***a* suitably fixed to a shaft **22**, the shaft **22'** being operable exteriorly of the tank **10**, by a crank **23***a* suitably geared to the shaft **22'**, as indicated at **23'**. With the valve slidably mounted before the aperture **24'** of the partition wall **20***a*, it will be apparent that the degree of opening of the aperture may be readily controlled with consequent control of flow therethrough.

Attention is now invited to Figure 7 of the drawings, wherein it will be seen that the propeller 21 comprises a hollow shell body 22, the hub 23 of which oscillatably support feathering blades 24, and in order to impart oscillatable movements to the blades, a shaft 25 is journalled, as at 26, in the shell 22. The shaft 25 includes a head 27 slotted to receive an arm 28 of each propeller 24.

An enlarged casing 29 is formed adapted for securement to the shell 22 and houses a plurality of weighted members 30, the members 30 being pivoted as at 31 to the casing, the inner ends of the weight members terminating in bill members 32 each of which is engageable in a kerf formed in the shaft 25. A helical spring 33 is interposed between the end of the shaft 25 and a thrust bearing 29′ slidably supported by the casing 29, exerting an endwise pressure upon the shaft tending to hold the shaft against rearward movement.

The casing 29 has integrally formed therewith an extension 34 connected to a drive shaft 35 by means of a universal joint 36.

From the description thus far described it will be understood that with rotation of the drive shaft 35 rotary motion will be imparted to the casing 22, and since the propellers 24 are supported by the casing 22, a similar motion will be transmitted thereto.

A housing 37 is provided for enclosing the casing 29 and affording a rockable bearing 38 for the casing 22, the rear portion of the housing having a flange 39 for mounting upon a wall of the tank 10. It will be apparent that the casing 22 is rockably mounted between the bearing 38 and the universal joint 36, so that the propeller hub 23 may be angularly adjusted to present the blades 24 at various degrees of inclination, from the horizontal axis of normal operation. The adjustment of the casing 22 is controlled exteriorly of the tank 10, as will now be set forth. A block 40 is disposed upon the extension 34, the block being of circular formation as seen in end elevation and upon the periphery thereof, a ring gear 41 is formed, and rearwardly of the ring gear, the block has threads 42 in mesh with internally formed threads of a supporting ring 43. The supporting ring 43 includes a pair of oppositely disposed projections or lugs adapted to engage within respective trackways 44 secured to the inner walls of the housing 37, and also has an ear 45 with which a lever 46 is pivotally connected. The lever 46 projects outwardly of the tank 10 and is supported intermediate its ends by a pin 47 engaged through a slot 48 extending longitudinally of the lever. It will be seen from the foregoing that the block 40 and associated propeller may be shifted upwardly or downwardly from a horizontal plane, by manipulation of the lever 46, the block 40 being guided by trackways 44.

The initial setting of the propeller blades 24 is effected by imparting a rotary motion to the block 40, which is carried out by a gear 49 in mesh with the ring gear 41. A suitable bearing 50 is formed upon the supporting ring 43, and has journalled therein a shaft 51 to which the gear 49 is keyed. The shaft 51 extends exteriorly of the tank 10, as shown and terminates in a hand wheel 52. Obviously, rotation of the shaft 51 will rotate the block 40 and since the block is in screw-threaded engagement with the ring 43 which is maintained against longitudinal movements, endwise movements will be transmitted to the shaft 25, through the medium of the thrust bearing 29′ and the helical spring 33. The drive for the propeller 21 is provided by the motor A, mounted upon a base B, the drive shaft C of which is journalled in a bearing D. The shaft 35 is also journalled in the bearing D and terminates in a thrust bearing E. The motor drive shaft C has keyed thereto a gear F which is in mesh with a similar gear G keyed to the shaft 35.

Obviously, the foregoing construction permits various settings of the propellers during operation, by merely rotating the block 40 through the hand wheel 52 and shaft 51, causing either a compression or permitting expansion of the spring. During compression of the spring an outward thrust would be imparted to the shaft 25, and since the arm 28 is connected with the head 27 a partial rotation on the longitudinal axis of the propeller would be effected, and upon expansion of the spring a reverse movement would be given the shaft 25 and propellers 24. It will also be noted that automatic adjustment of the propellers is permitted, which is effected through an increase or decrease of the speed of the motor A. The change in the speed of the motor may be varied manually or may be effected due to the varying density of the flow through the tunnel 19. An increase in the speed at which the shaft 35 is driven will cause the weights 30 to be swung outwardly by centrifugal force, developing a backward movement of the shaft 25, altering the setting of the blades, while a lowering of the speed will produce a forward movement of the shaft 25, providing a different setting of the blades 24. Therefore, the blades 24 will be set automatically in accordance with the speed of the shaft which will be commensurate with the fluidity of the material flowing through the tunnel 19.

If desired, in order to prevent ingress of water into the housing 37, around the rockable bearing 38, a rubber boot is provided, fixed to the bearing 38, by a ring 38*b*. The boot is of circular formation, entirely encircling the bearing 38 and includes a sleeve portion 38*c* encircling the casing 26*a* of the bearing 26. A clamping band 38*d* secures the sleeve upon the casing 26*a*.

From the foregoing, it will be apparent that water cannot enter the housing 37, yet the flexibility of the sleeve 38*c* will permit free movements of the rockable bearing 38.

Intermediate the length of the chute 16 an opening 53 is formed, controlled by a slide valve 54. The valve 54 comprises a rack bar 55 and a gear 56 in mesh therewith, the latter being operable exteriorly of the tank by a shaft and hand lever 57, and the purpose of this opening is in order that particles intermingled with the coal, which are of a buoyant nature, may seek egress therethrough and discharge into a semi-circular well 58 thereadjacent. Within the well 58 a pick-up drum 59 is revolubly mounted within bearings 60. The drum 59 includes a hub 61, comprising a plurality of circumferentially spaced bars 62 upon which there are secured in any approved manner, radially disposed perforated plates 63, extending the full length of the drum, the latter being slightly less in length than the inside length of the well, so that the drum will be readily accommodated within the well.

Upon each plate, at the outer extremities thereof, angle plates 64 are secured, positioned so that the angle plates will scoop up debris, as it leaves the opening 53 of the chute 16, and retain the same upon the plates 63 until the plates have reached a point directly above the chute 65 positioned centrally of and extending outwardly from the hub 61, when the debris will discharge into the chute. The chute 65 may discharge into a suitable container, or if desired, any well known conveyor may receive the debris from the chute.

The wall 66 defining the bottom wall of the well 58 is preferably grooved as at 67 to accommodate the angle bars 64 during rotation of the drum. It will be noted that the wall 66 stops short of the adjacent end wall of the tank, and has hingedly connected thereto a swinging valve plate 68. The plate 68 spans the space between the end of the wall 66 and the end wall of the tank, and adjustment for setting of the plate is accomplished through a pair of cables 69 trained around pulleys 70 fixed to a shaft 71. The shaft 71 is operable by a hand crank 72 exteriorly of the tank, and it will be obvious that the plate 68 may be set and held at various positions to increase or decrease the flow of water or passage of material into the drum 59.

The drive for the drum 59 will now be described and attention is invited particularly to Figs. 1, 2, and 6 of the drawings. The drum 59 has fixed to the ends thereof bevel gears 73 which mesh with respective gears 74 of a pair of shafts 75. The shafts 75 are suitably journalled in bearings 76 and have fixed thereto sprocket wheels 77. An electric motor 78 mounted upon a base 79 is provided, and includes a drive shaft 80. The shaft 80 has keyed thereon a sprocket wheel 81 aligned with one of the sprocket wheels 77 and around these two sprocket wheels there is trained a sprocket chain 82. Upon the base 79 there is also mounted a shaft 83 arranged parallel with the shaft 80 and fixed to the shaft there is a sprocket wheel 84 aligned with the other sprocket wheel 77, a sprocket chain 85 being trained about these sprocket wheels. In order to transmit power to the shafts 83 the shafts 80 and 83 have meshed gears 86 and 87 respectively. It will be seen from the foregoing that an even and positive drive is therefore provided for the drum.

In order to prevent heavy particles from reaching the propeller 21 a guard plate 88 is pivotally mounted upon a shaft 89, one end of which projects through the side wall of the tank 10 and has geared thereto, as at 90, a hand lever 91. The guard plate 88 may thus be adjusted by manipulation of the crank to vary the setting of the plate and stop lugs 92—93 limit the degree of opening and closing movements of the plate. In the full open position of the plate, as shown in dotted lines, it will be seen that any particles which have accumulated upon the plate 88 while in its closed or partially closed position will be deposited upon the inclined bottom wall 19*b* of the tank 10 leading to the bin 12, from whence they will be conducted to the conveyor 15 for removal, as will be presently described.

Immediately above the bin 12, a bin 95 is formed, within which a conveyor 96 is housed, the bin having communication at its upper side with the well 18, as indicated at 97. The conveyor 96 extends from the bin 95 to the outside of the tank 10 opposite that of the conveyor 15, as can be readily seen in Figures 1 and 4. The bin 95 has an inclined wall 98 extended upwardly at a slight angle to the lower wall of the chute 17 and also has an upper wall 99 which is in effect a continuation of the wall 66 of the well 58. The walls 98 and 99 are joined by an end wall 100 through which the chute 17 projects.

Extending longitudinally of the tank 10 in the medial plane thereof a tunnel 101 is suitably mounted, the underside of which is cut away, as at 102, forming communication with a bin 103. It will be noted from a consideration of Figure 2, that material discharging upwardly from the chute 17 will fall downwardly into the bin 103 to be removed by the conveyor 104, which extends to one side of the tank 10. The tunnel 101 is of greater diameter than the chutes 16 and 17, so as to allow ready passage of water and air therearound, as will be more fully explained in the description of the operation.

The bin 11 previously referred to is located directly below the bin 103 and receives sediment and small particles of debris which pass through the screened opening 105 of the bin 103. The sediment passing from the opening will fall or settle downwardly upon the conveyor 14, guided by the inclined wall 106. The conveyor 14 extends to one side of the tank 10, opposite that of the conveyor 104.

Within the forward end of the tunnel 101 I provide a feathering propeller 107, the blades 108 of which are adjustable to varying settings through rotation of the hollow shaft 109 by the hand wheel 110. The shaft 109 is threaded as at 109*a* cooperable with a threaded portion of a bearing 111 on the tank 10, and by bearing 112 mounted within the tunnel 101. Any suitable lubricating means may be provided and in the present instance, I have illustrated a grease cup 113 associated with the bearing 112. A drive shaft 114 extends through the hollow shaft 109 exteriorly of the tank, journalled as at 115, and terminates in a thrust bearing 116. It will be obvious that upon rotation of the shaft 109 by the hand wheel 110, the shaft 109 will be moved longitudinally with respect to the drive shaft 114, by virtue of the threaded engagement between the bearing 111 and the shaft 109. As the propeller 107 is rotatably mounted by the shaft 109 and movable longitudinally therewith, the propeller blades 108 will be rotated due to their engagement with the headed portion of the drive shaft 114. A motor 117 suitably supported by a base 118 is provided for imparting power to the shaft 114 and includes a drive shaft 119, the outer end of which is supported in the journal 115. A gear 120 is keyed to the shaft 119, which is in mesh with a gear 121 keyed to the shaft 114.

From the foregoing, it will be clearly apparent that rotation of the propeller 107 may be effected upon starting of the motor 117 and that the setting of the blades 108 may be accomplished while the propeller is in operation, thereby increasing or decreasing the effectiveness of the propeller, as conditions demand.

A screen 122 may be installed in the tunnel 101 in advance of the propeller 107 to prevent the passage of débris into the tunnel and to also diffuse the mixture of air and water as will be dealt with hereinafter.

A vertically disposed baffle wall 123 extends transversely of the tank 10 and is secured to the wall 98 at its upper end, the lower end being deflected as at 124 and stops short of the bottom wall of the bin 12, and to one side of the conveyor 15, so that material entering the bin 12 will be shunted toward the conveyor 15, for removal from the bin.

A wall 125 is built into the bin 11 extending parallel with the wall 123, and stops short of the wall 98, and it should also be noted that a portion of the bottom wall of the bin 12 extends parallel with the deflector wall 124, as indicated at 126. Thus, the walls 123, 125 and 126 and deflector 124 define a passageway 127 for circulation of water in the bins 11 and 12.

It has been found in practice that sometimes the bins 11 and 12 become clogged with sediment which is not removed by their respective conveyors, and provision is therefore made for removal of such sediment. Associated with the bin 11 I have illustrated a pair of superposed sliding shutters 128 and 129. These shutters 128—129 are slidable in trackways 130—131 respectively, built into the chute framing 132, and may include any suitable means, such as the hand wheel 133 for sliding the shutters backwardly and forwardly to control the passage of sediment through the aperture 134 of the chute. In use, the shutters are normally in closed position, and when it is desired to clear the bin 11 of sediment, the uppermost shutter 128 is drawn outwardly so as to clear the aperture 134. The sediment will then be free to flow through the aperture 134 and will be deposited upon the shutter 129. The shutter 128 is then closed and the shutter 129 is opened, when the sediment may flow through the funnel portion 135 of the chute into a suitable receptacle or otherwise.

A single shutter 136 has been shown as associated with the bin 12, but obviously the superposed arrangement described in connection with the bin 11 may be employed, if desired.

With further reference to the chutes 16 and 17 these chutes are lined as indicated at 136 to avoid wear upon the chutes proper, due to passage of material therethrough. The chute 17 also includes a sliding valve door 137 operable to open and close an aperture 138 formed in the lower side of the chute. Any suitable means may be employed for operating the door and in the present instance, I have illustrated the door as having a rack bar 139 fixed to the back thereof and in mesh with the rack bar, there is a gear 140. The gear 140 is keyed to a shaft 141 extending transversely of the tank 10 and projecting through one side thereof, and carries a ratchet gear 142. A pawl 143 is pivotally mounted upon the tank and operable to engage the ratchet gear to maintain the shaft against rotation. A hand lever 144 is also fixed to the shaft 141, providing means for manually rotating the shaft 141 to vary the setting of the door 137.

The tank 10 may be supported upon any suitable base, but in the present instance, I have illustrated a base framing 145, upon which there are erected upright support members 146 of varying heights for support of troughs 147—148—149—150 of the conveyors 14—15—96 and 104 respectively. The conveyor troughs 147—150 may have any suitable inclination, for the most efficient operation of the conveyors.

The operation will be readily understood from a consideration of the following description thereof. The tank 10 is filled with water to a height indicated by the water-line W, and the motors A, 78 and 117 are set in motion. The water in the tank will be circulated therearound by virtue of the propellers 21 and 107, and since the tunnel 101 is partially open to the atmosphere, the propeller 107 will draw air into the tunnel and there is mixed by the propeller 107 with the water in the tunnel. The mixing of the air and water will effect a final clearing of the coal emerging from the chute 17, as will be dealt with hereinafter.

Coal to be cleaned is fed into the chute 16 settling downwardly into the water toward the well 18. This coal as it comes from the mine culm bank and streams contains considerable quantities of cinders, wood chips and similar foreign matter of a floatable character, and since the chute 16 is filled with water above the opening 53 formed therein is partly submerged, the floating material will pass through the opening 53 into the well 58 where the material is then picked up by the plates 63 of the drum 59. The plates 63 will maintain the débris thereon until they reach a point directly above the chute 65 where the material will slide from the plates into the chute 65, where it may be disposed of, as desired.

The coal entering the chute 16 will also contain a considerable amount of slate, which being heavier than the coal, a large part of the slate will fall through the port 97 into the bin 95 to be removed by the conveyor 96. It should be noted that as the coal and slate pass from the chute 16 to the well 18, they encounter a cross spiral flow of water developed by the propeller 21, as indicated by the arrows in Figure 2. The blades 24 of the propeller are given a set adjacent their ends opposite to that next the hub, so that the water in the chute 17 will be caused to flow in two distinct paths, namely the water adjacent the inside surfaces of the chute will be caused to flow outwardly toward the discharge end 17′ of the chute, while the water centrally of the chute will be drawn toward the propeller 21. Such a cross flow of water will cause a high agitation of the water or a "push" or "pull" and "roll" action upon the coal and slate tending to thoroughly separate the coal from slate into a more or less buoyant aggregate, permitting the slate which may still be mixed with the coal to pass through the opening 138 formed in the lower side of the chute 17. The slate passing through the latter opening will fall upon the inclined wall 98 and will be shunted to the conveyor bin 95 where it will be removed by the conveyor 96.

The coal in the chute 17 will be floated over the discharge end 17′ where it will encounter a mixture of air and water developed by the propeller 107. The mixture of air and water will tend to further cleanse the coal of foreign matter, the latter being forced through the tunnel, around the chutes 16 and 17 and finally settling in the drum 59.

The finally cleaned coal settles upon the conveyor 104 to be removed from the tank. Obviously, considerable fine sediment will be mixed with the water, and such sediment will follow circulation of water through the screen 105 whereafter the flow of water occupies a greater area losing velocity in recirculation, allowing sediment to settle upon the conveyor 14, permitting its removal.

The circulation of the water in the tank 10 will be substantially as follows: through the tunnel 101 into the well 58, the exit therefrom being controlled by the settling of the valve door 68, through the chamber 19*a*, the tunnel 19; the chute 17, passing therefrom into the well 103 and thence into the bin 11. The water finally entering the chamber 19*a* by way of the passageway 127, the bin 12 and opening 24′ for recirculation.

A certain amount of débris will come to rest upon the door 68, and this may be removed by allowing the door to swing to full open position by manipulation of the hand crank 72 depositing the débris upon the plate 88. As has been previously indicated, the plate 88 is operable by virtue of the lever 91 to discharge the débris upon the inclined floor 19b from whence it will pass to the conveyor 15 for removal.

After continuous operation of the apparatus for any considerable length of time, it is obvious that the water in the tank will become quite dense from slush sediment forming a more solid body in water current, causing a greater buoyancy in spiral current of chute 17. The push agitation against the water must then be less, which is caused automatically by density of the water which is slowing up the revolutions of motor A controls the centrifugal action of the weight members 30, which automatically controls adjustment of the propeller blades 24.

The push agitation against the water is caused by part 21w of the blades and the pull agitation of the water is effected by part 21x of the blades and the whirl or roll agitation of the water is produced by the centrifugal action of the revolving propeller, as an entirety.

During operation of the apparatus, in order to further insure floating of cinders on the surface of the water, I distribute ground cork on the surface of the water, forming a recirculating buoyant net moving continuously with the current formed by the propeller 107. It should be understood that the cinders have previously passed through the well 18 where they have been thoroughly cleaned of culm, by the spiral current of water, leaving the pores or crevices of the cinders open, permitting the entrance of particles of ground cork thereinto. Thus the buoyancy of the cinders is greatly increased.

From the foregoing, it will be seen that I have provided an apparatus which will effectively separate débris from coal in a highly efficient manner, requiring a minimum of labor, and while I have shown and described a preferred embodiment of the invention, I do not confine myself to the exact construction shown, and reserve as my own, all such modifications as fairly fall within the scope of the appended claims.

I claim:—

1. In a coal and slate separating apparatus, a tank containing water, a well submerged in the water, said well including a feed chute and a discharge chute, said discharge chute stopping slightly below the water level of the tank, propeller means for circulating a mixture of air and water across the mouth of the discharge chute whereby to separate floatable particles from the coal issuing from the chute, means for removing the floating particles, a bin formed beneath the discharge chute, conveyor means within the bin for removing coal discharged into the bin, the well having a discharge opening in the base thereof, conveyor means therebeneath for removing material discharged therethrough, water inlet means formed in the well establishing communication with the water in the tank, and means for creating a simultaneous oppositely flowing current of water through the well and discharge chute.

2. In a coal and slate separating apparatus, a tank containing water, a well submerged in the water, said well including a feed chute and a discharge chute, a revolving pick-up drum mounted within the tank in proximity to the feed chute, said feed chute having a port in communication with the drum; said discharge chute being submerged slightly below the water level of the tank, propeller means for circulating a mixture of air and water across the mouth of the discharge chute whereby to separate the floatable particles from the coal issuing from the chute and carry the same to the pick-up drum, a bin formed beneath the discharge chute, conveyor means within the bin for removing coal discharged into the bin, the well having a discharge opening formed in the base thereof, conveyor means therebeneath for removing material discharged therethrough, water inlet means formed in the well establishing communication with the water in the tank, and propeller means for creating a simultaneously oppositely flowing current of water through the well and discharge chute.

3. In a coal and slate separating apparatus, a tank containing water, a well submerged in the water, said well including a feed chute and a discharge chute, a revolving pick-up drum mounted within the tank in proximity to the feed chute, a port formed in the feed chute in communication with the drum; said discharge chute stopping slightly below the water level of the tank, propeller means for circulating a mixture of air and water across the mouth of the discharge chute whereby to separate the floatable particles from the coal issuing from the chute and carry the same to the pick-up drum, a bin formed beneath the discharge chute, conveyor means within the bin for removing coal discharged into the bin, the well having a discharge opening in the base thereof, conveyor means therebeneath for removing material discharged therethrough, water inlet means formed in the well establishing communication with the water in the tank, and automatic adjustable propeller means for creating simultaneously oppositely flowing currents of water through the well and chute.

4. In a coal and slate separating apparatus, a tank containing water, a pair of bins formed in the base thereof, partition walls between the bins defining a passage-way therebetween for circulation of water from one bin to the other, conveyors positioned within the bins for removing material deposited within the bins, a well submerged in the water, said well including a feed chute and a discharge chute, a revolving pick-up drum mounted within the tank in proximity to the feed chute, a regulatable port in the feed chute in communication with the drum for permitting passage of débris into the drum; said discharge chute stopping short slightly below the water level of the tank, adjustable propeller means for circulating a mixture of air and water across the mouth of the discharge chute whereby to separate the floatable particles from the coal issuing from the discharge chute and carry the same to the pick-up drum, a bin formed beneath the discharge chute, conveyor means within the bin for removing coal discharged into the bin; the well having a discharge opening in the base thereof, a bin formed therebeneath, conveyor means within said bin, the bin having a well extending substantially in the plane of the lower wall of the discharge chute defining a passageway communicating with the last named bin, a regulatable port in the discharge chute forming communication with the last named passageway, water inlet means formed in the well establishing communication with the water in the tank, and automatic adjustable propeller means for creating a simultaneously oppositely flowing current of water through the well and discharge chute.

5. In a coal and slate separating apparatus, a tank containing water, a pair of bins formed in the base thereof, partition walls between the bins defining a passage-way therebetween for circulation of water from one bin to the other, con-

veyor means in the base of the bins for removing sediment deposited therein, valve means in the bins below said conveyors for withdrawing the fluid sediment in the bins not removed by said conveyors, a well within the tank and submerged in the water, said well including a feed chute and a discharge chute, the feed chute opening upon the upper side of the tank, a tunnel extended longitudinally of the tank and partly submerged in the water therein, the mouth of the discharge chute opening into said tunnel and below the water line in the tunnel, variable propulsion means in said tunnel in advance of the mouth of the discharge chute for agitating and mixing air and water in the tunnel, a bin positioned beneath the mouth of the discharge chute, a conveyor in the bin for removing coal from the bin, said bin having an overflow port in communication with one of the bins of the tank, a revoluble pick-up drum mounted within the tank and receiving débris from the tunnel and the feed chute, a chute mounted within the drum and receiving the débris picked up thereby, a compartment formed adjacent one of the bins of the tank, means for controlling flow of water through the compartment and the bin, means controlling flow of water and débris from the drum and the compartment, a port between the well and the compartment, automatic adjustable propeller means positioned at the port for creating a simultaneously oppositely flowing current of water through the aggregate fed through the feed chute, and means in the base of the well to remove slate settled out of the aggregate, said propeller means further creating a circulation of water throughout the tank for deposition of material into the various bins.

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